

TASK S5.01

CONSULTAS MONGODB

Descripció

Treballarem amb una base de dades que conté col·leccions relacionades amb una aplicació d'entreteniment cinematogràfic:

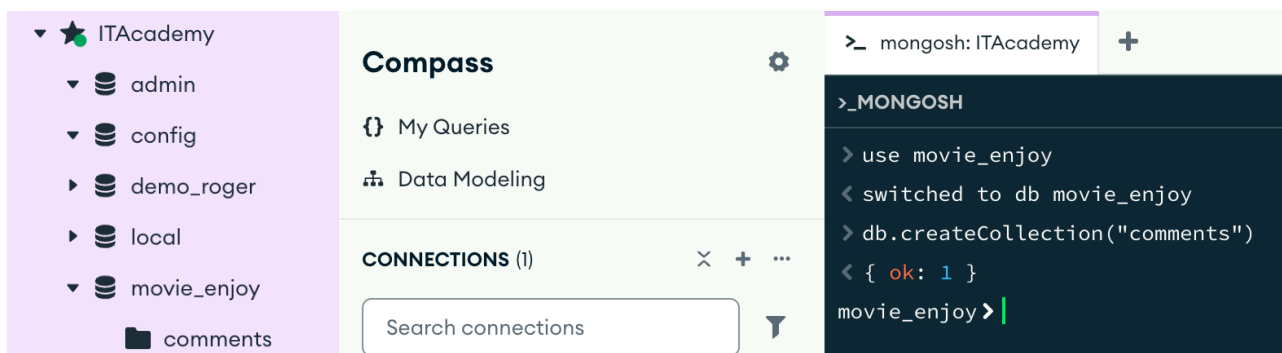
- users: Emmagatzema informació d'usuaris/es, incloent-hi noms, emails i contrasenyes xifrades.
- theatres: Conté dades de cinemes, com ID, ubicació (direcció i coordenades geogràfiques).
- sessions: Guarda sessions d'usuari, incloent-hi ID d'usuari i tokens JWT per a l'autenticació.
- movies: Inclou detalls de pel·lícules, com a trama, gèneres, durada, elenc, comentaris, any de llançament, directors, classificació i premis.
- comments: Emmagatzema comentaris d'usuaris/es sobre pel·lícules, amb informació de l'autor/a del comentari, ID de la pel·lícula, text del comentari i la data.

Duràs a terme algunes consultes que et demana el client/a, el qual està mesurant si seràs capaç o no de fer-te càrrec de la part analítica del projecte vinculat amb la seva base de dades.

Level 1

Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.

Level 1 - Database creation and initialization



For the import of the Json comments I have used the command including 'collection' because I have already created the collection in my db of mongoDB

Mongoimport --db movie_enjoy --collection comments --file "comments.json"

For movies.json and the rest of Json, I use the advantage that Mongoimport creates the collection in the DB in case it is not indicated --collection

Mongoimport --db movie_enjoy --file "movies.json"

```

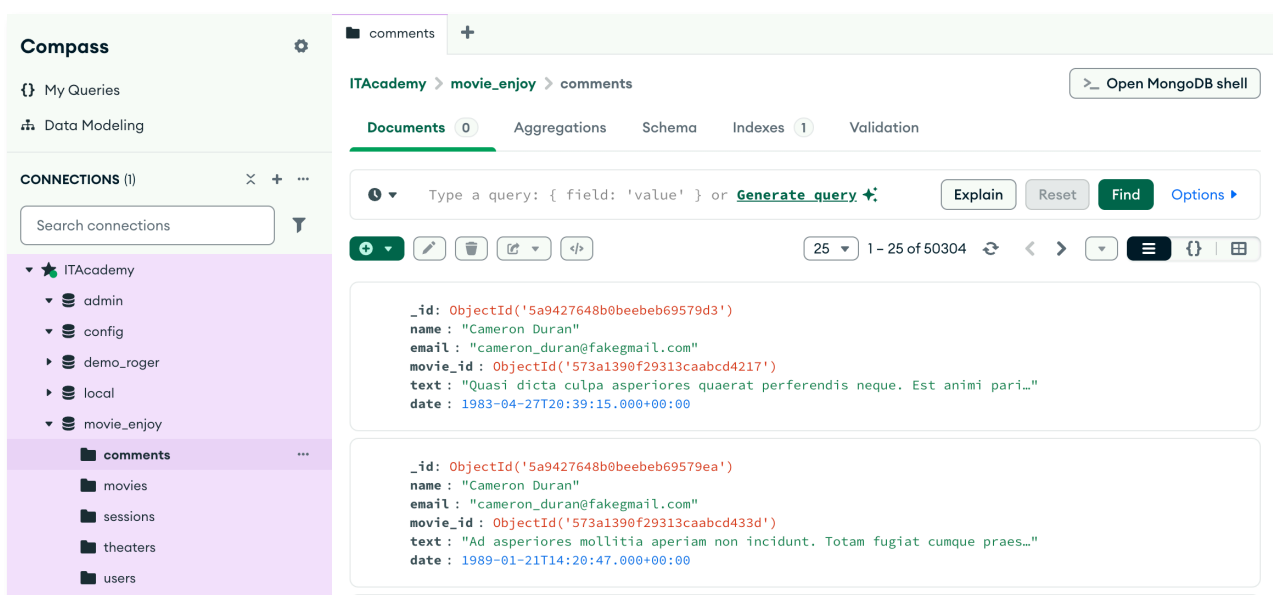
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --collection comments --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/comments.json'
2026-03-05T20:33:48.558+0100 connected to: mongodb://localhost/
2026-03-05T20:33:49.270+0100 50304 document(s) imported successfully. 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/movies.json'
2026-03-05T20:49:18.125+0100 no collection specified
2026-03-05T20:49:18.127+0100 using filename 'movies' as collection
2026-03-05T20:49:18.136+0100 connected to: mongodb://localhost/
2026-03-05T20:49:19.095+0100 23539 document(s) imported successfully. 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/sessions.json'
2026-03-05T20:50:36.430+0100 no collection specified
2026-03-05T20:50:36.431+0100 using filename 'sessions' as collection
2026-03-05T20:50:36.438+0100 connected to: mongodb://localhost/
2026-03-05T20:50:36.462+0100 1 document(s) imported successfully. 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/theaters.json'
2026-03-05T20:52:05.709+0100 no collection specified
2026-03-05T20:52:05.710+0100 using filename 'theaters' as collection
2026-03-05T20:52:05.716+0100 connected to: mongodb://localhost/
2026-03-05T20:52:05.752+0100 1564 document(s) imported successfully. 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ % mongoimport --db movie_enjoy --file '/Users/rogerdefez/Documents/Cursos i Llibres/BootCamp IT Academy/03_Especialitzacio/Sprint_5_DB_No_Relacionals/Dades originals_S5/users.json'
2026-03-05T20:52:56.529+0100 no collection specified
2026-03-05T20:52:56.530+0100 using filename 'users' as collection
2026-03-05T20:52:56.539+0100 connected to: mongodb://localhost/
2026-03-05T20:52:56.567+0100 185 document(s) imported successfully. 0 document(s) failed to import.
rogerdefez@MacBook-Air-de-Roger ~ %

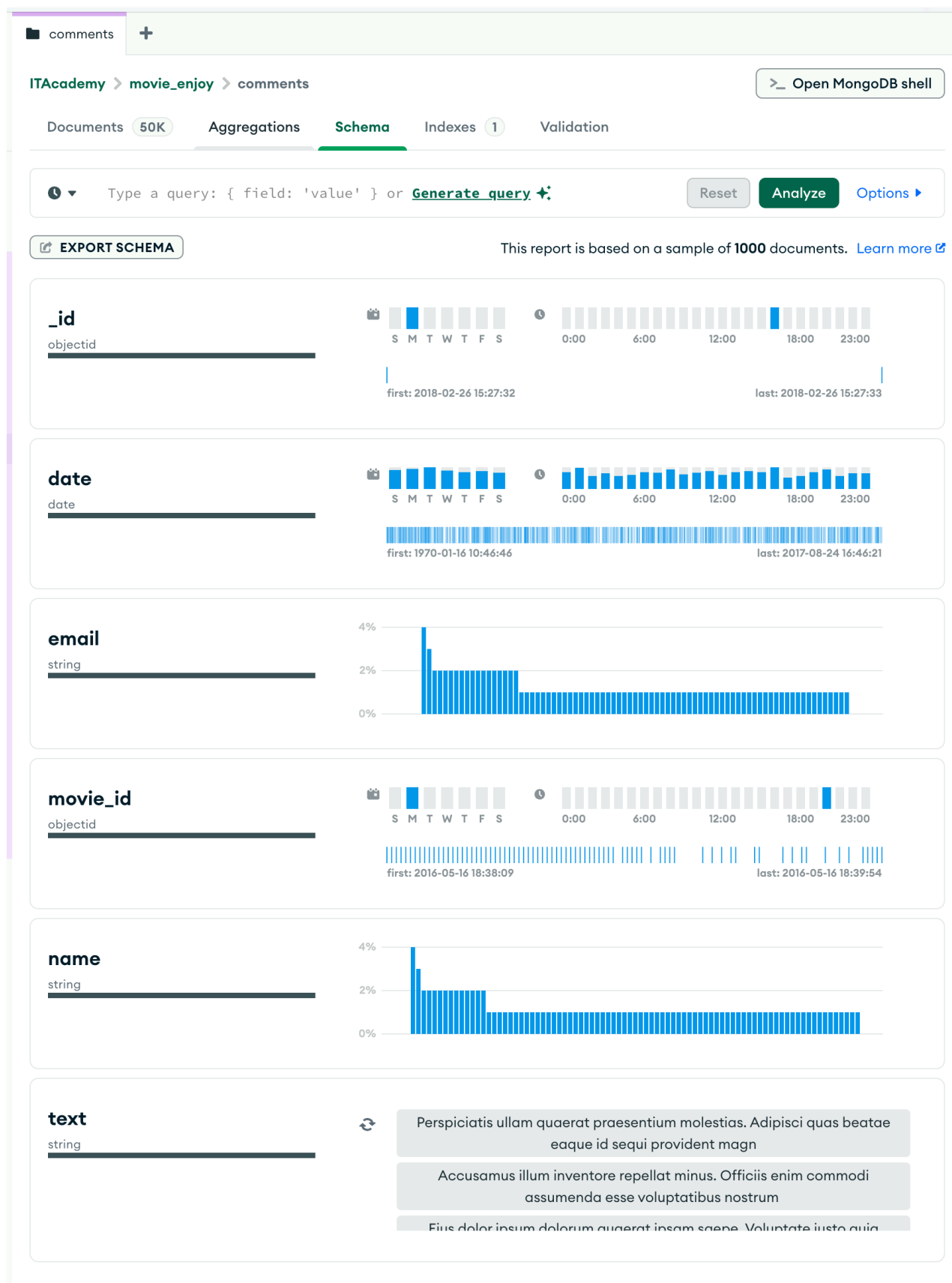
```

Analysis EDA

An important step before continuing with the Task 5 queries, it is important to do an EDA - Exploratory Data Analysis.

Below I attach screenshots of the first documents of each collection.





The screenshot displays the MongoDB Compass web interface. On the left, the 'Compass' sidebar shows a tree of connections under 'ITAcademy', with 'movie_enjoy' selected. The main panel shows the 'movies' collection with 24K documents. A document is selected, showing its JSON structure. The document includes fields like '_id', 'plot', 'genres', 'runtime', 'rated', 'cast', 'num_mflix_comments', 'poster', 'title', 'fullplot', 'languages', 'released', 'directors', 'writers', 'awards', 'lastupdated', 'year', 'imdb', 'countries', 'type', and 'tomatoes'.

Compass

My Queries
Data Modeling

CONNECTIONS (1)

Search connections

ITAcademy

- admin
- config
- demo_roger
- local
- movie_enjoy
 - comments
 - movies**
 - sessions
 - theaters
 - users
- sample_airbnb
- sample_analytics
- sample_geospatial
- sample_guides

movies

ITAcademy > movie_enjoy > movies

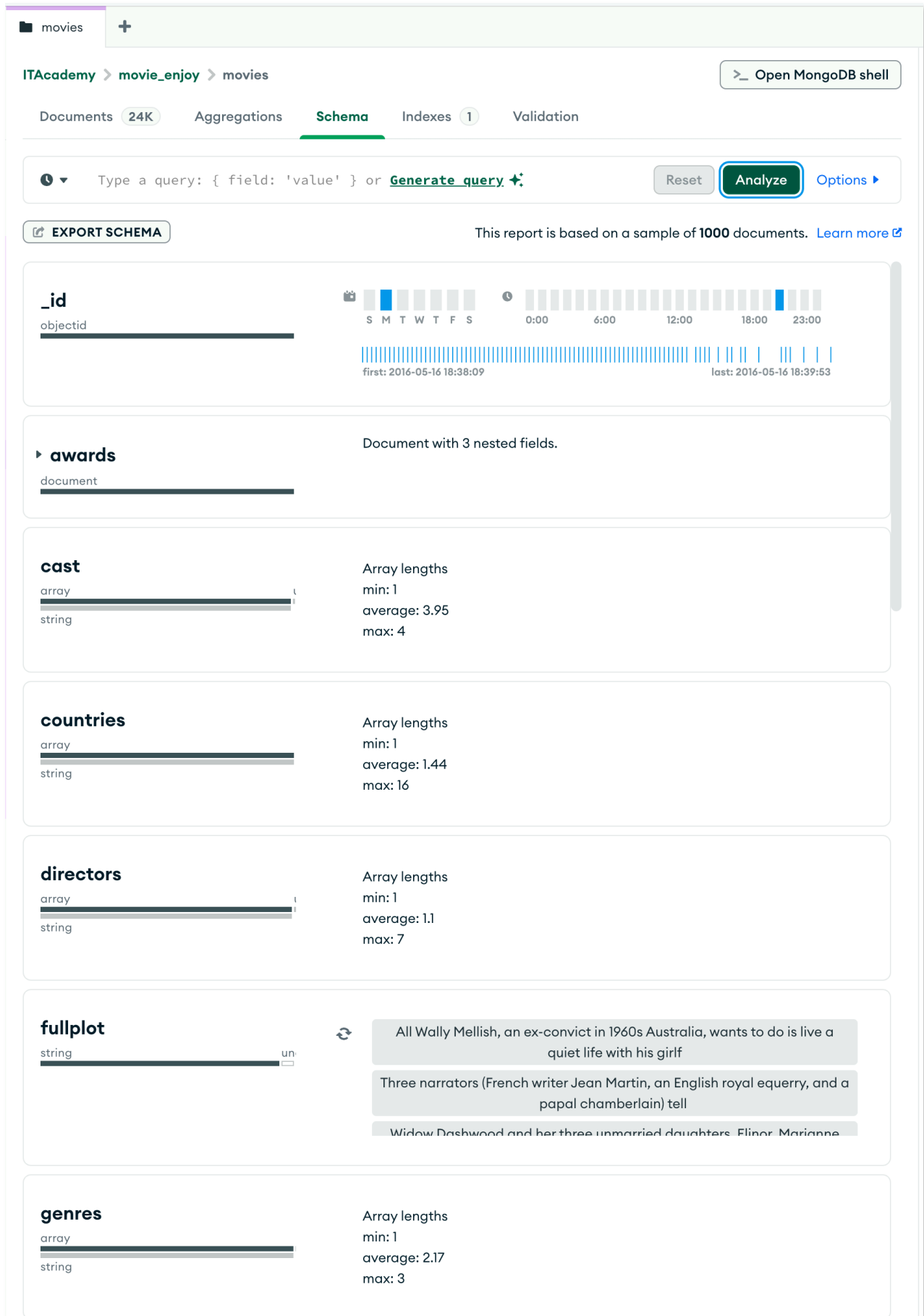
Documents 24K Aggregations Schema Indexes 1 Validation

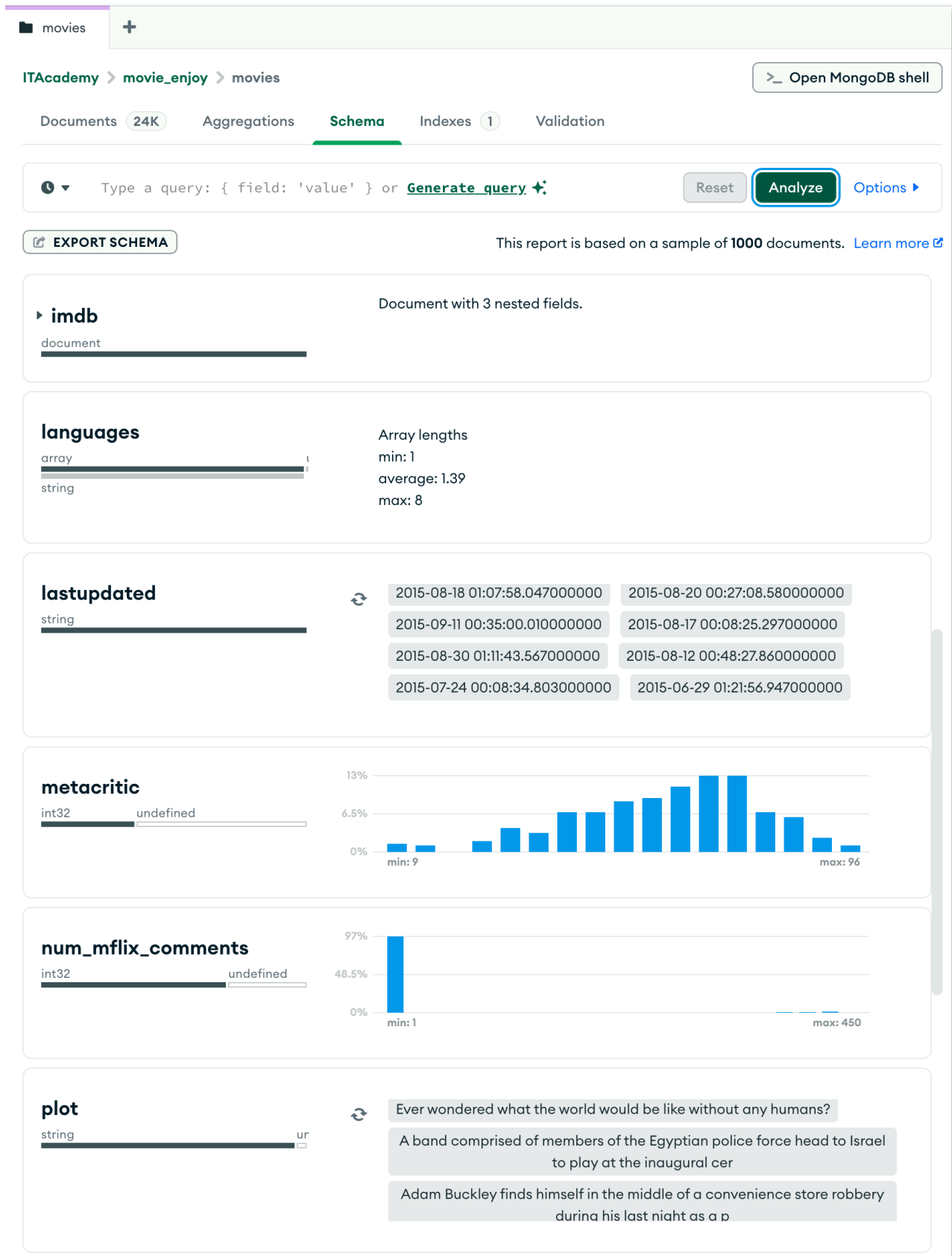
Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

25 1 - 25 of 23539

```
{
  "_id": ObjectId('573a1390f29313caabcd4323'),
  "plot": "A young boy, oppressed by his mother, goes on an outing in the country ...",
  "genres": Array (3),
  "runtime": 14,
  "rated": "UNRATED",
  "cast": Array (4),
  "num_mflix_comments": 2,
  "poster": "https://m.media-amazon.com/images/M/MV5BMTMzMjQxMjgyNl5BMTU5BnBnXkFtZT...",
  "title": "The Land Beyond the Sunset",
  "fullplot": "Thanks to the Fresh Air Fund, a slum child escapes his drunken mother ...",
  "languages": Array (1),
  "released": 1912-10-28T00:00:00.000+00:00,
  "directors": Array (1),
  "writers": Array (1),
  "awards": Object,
  "lastupdated": "2015-08-29 00:27:45.437000000",
  "year": 1912,
  "imdb": Object,
  "countries": Array (1),
  "type": "movie",
  "tomatoes": Object
}
```



movies

+

ITAcademy > movie_enjoy > movies

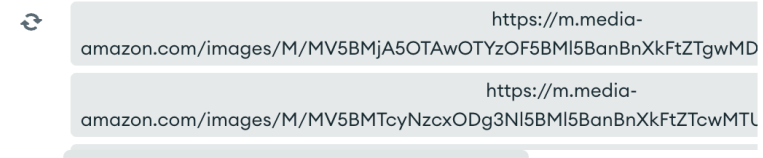
Documents 24K Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) [Reset](#) [Analyze](#) [Options](#)

[EXPORT SCHEMA](#) This report is based on a sample of 1000 documents. [Learn more](#)

poster

string undefined

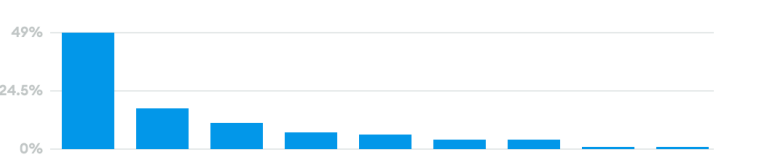


https://m.media-amazon.com/images/M/MV5BMjA5OTAwOTYzOF5BMl5BanBnXkFtZTgwMD

https://m.media-amazon.com/images/M/MV5BMjA5OTAwOTYzOF5BMl5BanBnXkFtZTgwMD

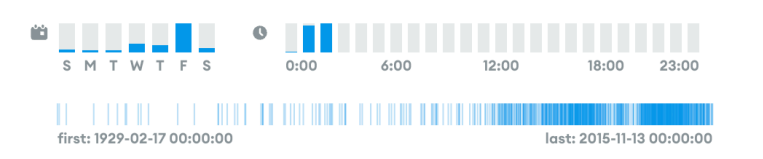
rated

string undefined



released

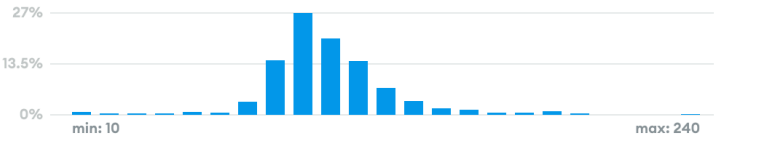
date



first: 1929-02-17 00:00:00 last: 2015-11-13 00:00:00

runtime

int32



min: 10 max: 240

title

string



Beverly Hills Cop III The Grave The Scarlet Letter

She's Out of Control Not Another Happy Ending Red Sands

Kuffs Clerks Adventures in the Sin Bin

20th Century Boys 2: The Last Hope A Knight's Tale Floating

tomatoes

document

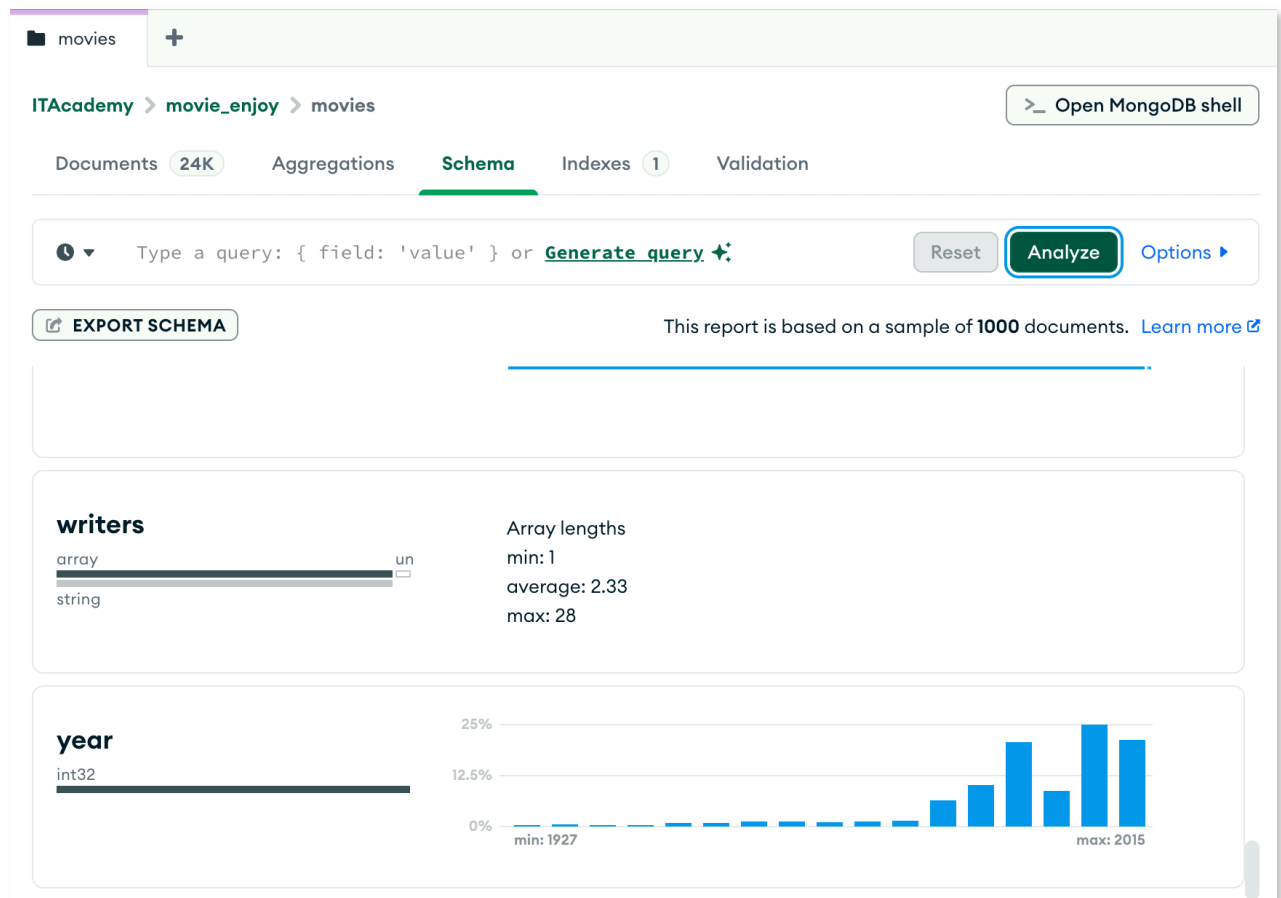
Document with 10 nested fields.

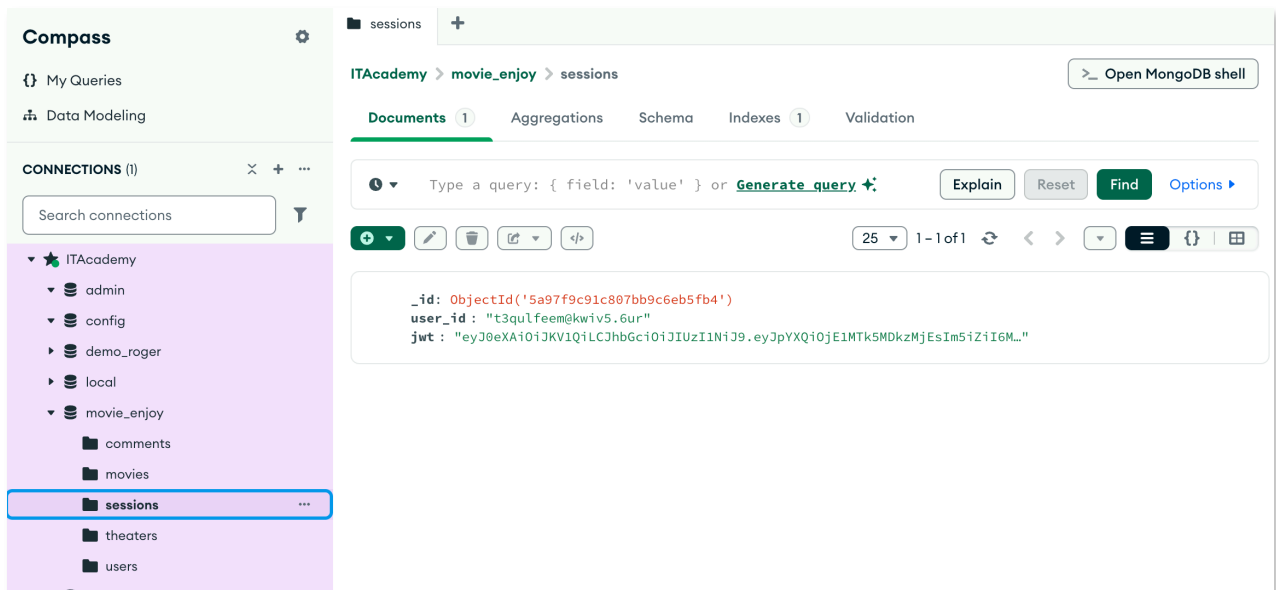
type

string



movie





Compass

My Queries
Data Modeling

CONNECTIONS (1)

Search connections

ITAcademy

- admin
- config
- demo_roger
- local
- movie_enjoy
 - comments
 - movies
 - sessions**
 - theaters
 - users

sessions +

ITAcademy > movie_enjoy > sessions

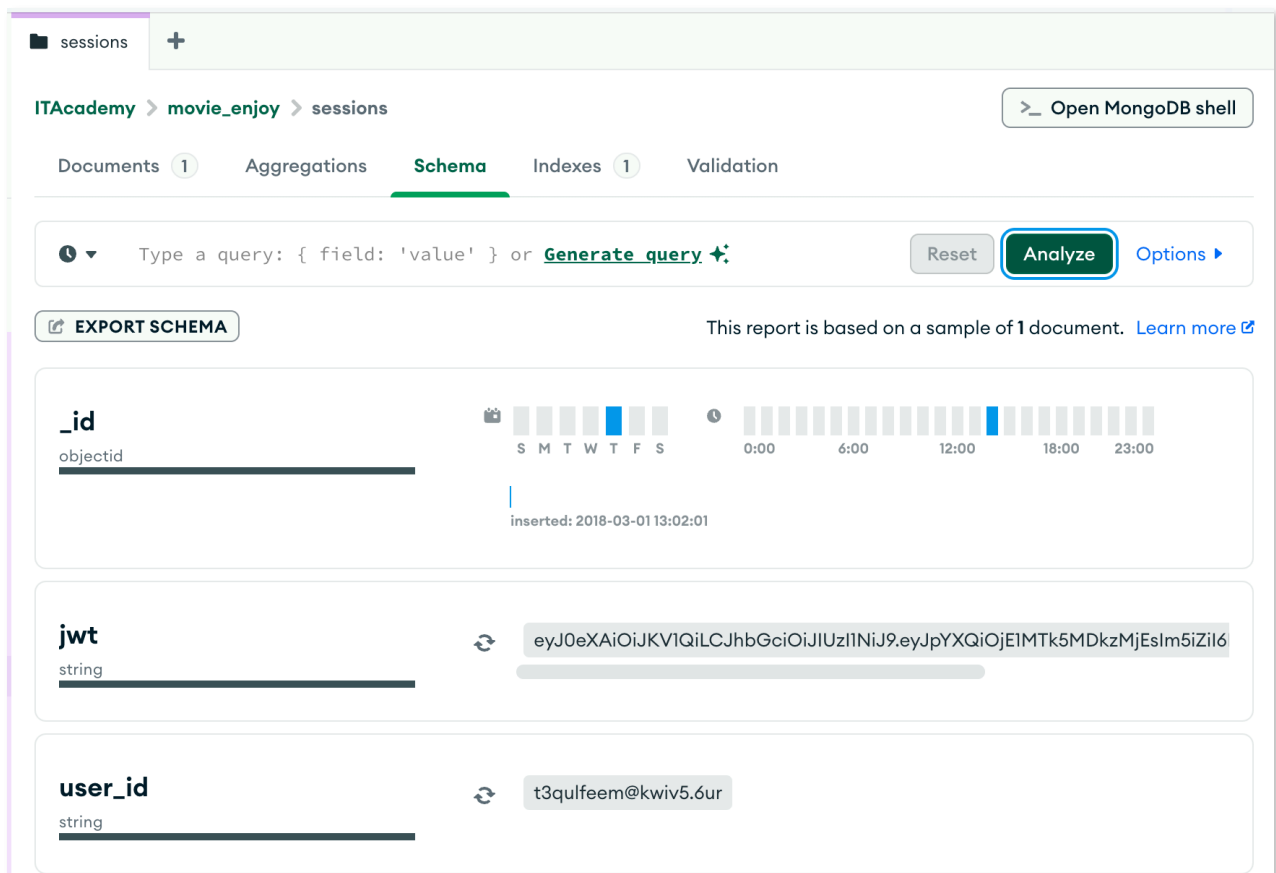
Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

25 1 - 1 of 1

```
{
  "_id": ObjectId('5a97f9c91c807bb9c6eb5fb4'),
  "user_id": "t3qulfeem@kwiv5.6ur",
  "jwt": "eyJ0eXAiOiJKVlQiLCJhbGciOiJIUzI1NiJ9.eyJpYXQiOiJlMTk5MDkzMjEsIm5iZiI6I6M..."
}
```



sessions +

ITAcademy > movie_enjoy > sessions

Documents 1 Aggregations **Schema** Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Reset Analyze Options

[EXPORT SCHEMA](#)

This report is based on a sample of 1 document. [Learn more](#)

_id
objectid

S M T W T F S 0:00 6:00 12:00 18:00 23:00

inserted: 2018-03-01 13:02:01

jwt
string

eyJ0eXAiOiJKVlQiLCJhbGciOiJIUzI1NiJ9.eyJpYXQiOiJlMTk5MDkzMjEsIm5iZiI6I6M...

user_id
string

t3qulfeem@kwiv5.6ur

Compass

{} My Queries

Data Modeling

CONNECTIONS (1)

Search connections

ITAcademy

- admin
- config
- demo_roger
- local
- movie_enjoy
 - comments
 - movies
 - sessions
 - theaters**
 - users

theaters

ITAcademy > movie_enjoy > theaters

Documents 1.6K Aggregations Schema Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN Reset Find Options

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE

25 1 - 25 of 1564

```
{
  "_id": ObjectId("59a47286cfa9a3a73e51e72e"),
  "theaterId": 1008,
  "location": {
    "address": {
      "street1": "1621 E Monte Vista Ave",
      "city": "Vacaville",
      "state": "CA",
      "zipcode": "95688"
    },
    "geo": {
      "type": "Point",
      "coordinates": [
        -121.96328,
        38.367649
      ]
    }
  }
}
```

theaters

ITAcademy > movie_enjoy > theaters

Documents 1.6K Aggregations **Schema** Indexes 2 Validation

Type a query: { field: 'value' } or [Generate query](#)

Reset Analyze Options

EXPORT SCHEMA

This report is based on a sample of 1000 documents. [Learn more](#)

_id
objectid

S M T W T F S 0:00 6:00 12:00 18:00 23:00

first: 2017-08-28 19:44:06 last: 2017-08-28 19:44:07

location
document

Document with 2 nested fields.

address
document

Document with 5 nested fields.

city
string

3% 1.5% 0% 0:00 6:00 12:00 18:00 23:00

state
string

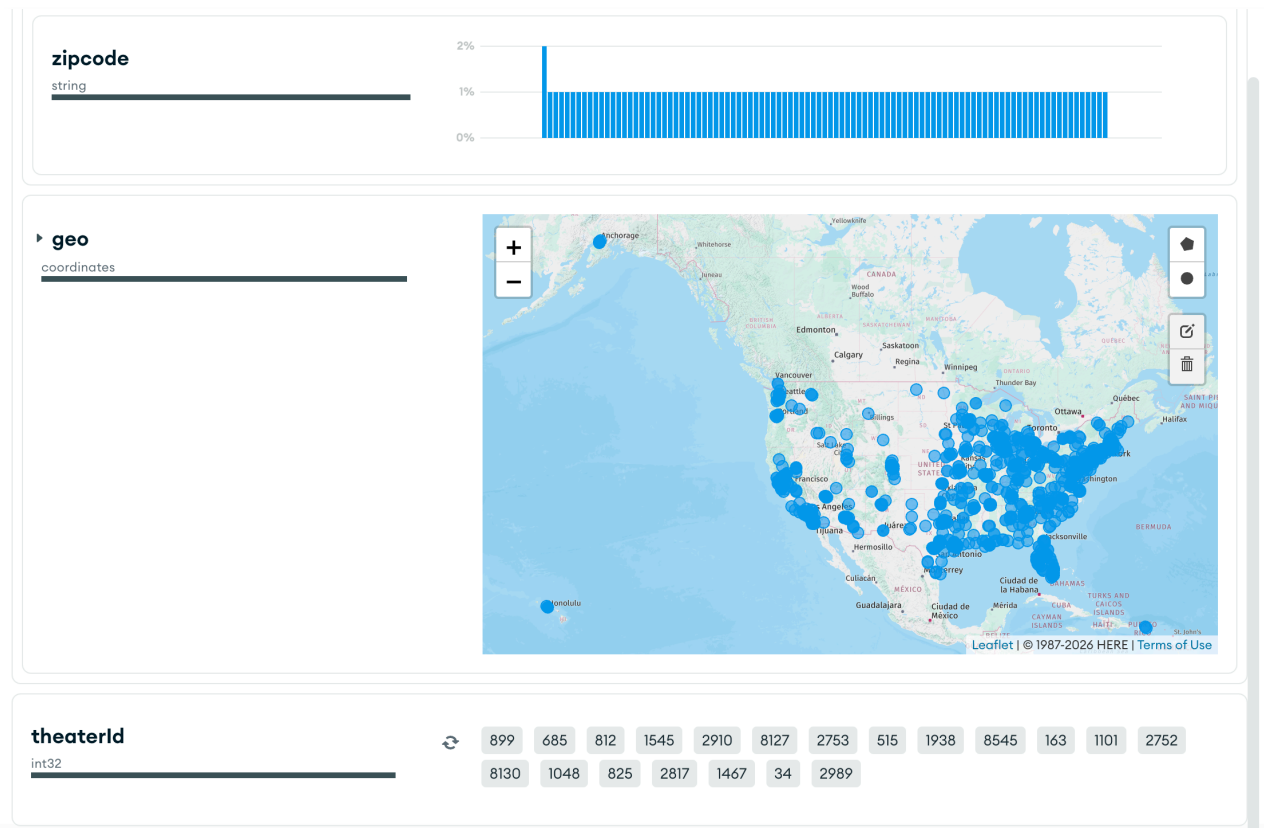
22% 11% 0% 0:00 6:00 12:00 18:00 23:00

street1
string

2510 Gulf Fwy S 3751 32nd Ave S 5000 West International Airport Road
1100 Oxford Exchange Blvd 120 Slater St 35000 Warren Rd. 7601 Penn Ave South
1501 N Victory Pl 175 Grandview Blvd 3675 Pacific Coast Hwy 330 Connecticut Ave
310 Daniel Webster Hwy 2003 N Loop 1604 E 1201 Boston Post Road 5890 N Elizabeth St

street2
string null undefined

2% 1% 0% 0:00 6:00 12:00 18:00 23:00



Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

ITAcademy

admin

config

demo_roger

local

movie_enjoy

comments

movies

sessions

theaters

users

sample_airbnb

users

+

ITAcademy > movie_enjoy > users

Documents 185

Aggregations

Schema

Indexes 1

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

25

1 - 25 of 185

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25

1 - 25 of 185

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⌵

_id: ObjectId('59b99db9cfa9a34dcd7885bf')

name: "Jon Snow"

email: "kit_harington@gameofthron.es"

password: "\$2b\$12\$fDEu1Ru66tLWAVidMN.b0.929B1fnyqdGuhWMyzf0Af/ATY0yLoY6"

_id: ObjectId('59b99db9cfa9a34dcd7885c0')

name: "Sansa Stark"

email: "sophie_turner@gameofthron.es"

password: "\$2b\$12\$nCIVE81..AtAoySPZkl19.G5V0EdIwwSh1f18lxWEr3d1pG/Uusi"

_id: ObjectId('59b99dbacfa9a34dcd7885c1')

name: "Arya Stark"

email: "maisie_williams@gameofthron.es"

password: "\$2b\$12\$19h3KjeTg3/sPNFHztdb6eGWkrCxIwlVXVSh9POSi5PS9kSlguZEq"

users

+

ITAcademy > movie_enjoy > users

>_ Open MongoDB shell

Documents 185AggregationsSchemaIndexes 1Validation

Type a query: { field: 'value' } or [Generate query](#)

ResetAnalyzeOptions

EXPORT SCHEMA

This report is based on a sample of 185 documents. [Learn more](#)

_id

objectid

SMTWTFSS

0:006:0012:0018:0023:00

first: 2017-09-13 21:05:56last: 2019-10-24 15:30:06

email

string

justin_williams@fakegmail.com

april_cole@fakegmail.com

christian_williams@fakegmail.com

hafthór_júlíus_björnsson@gameofthron.es

gwendoline_christie@gameofthron.es

yolanda_owen@fakegmail.com

kerry_ingram@gameofthron.es

jose_hall@fakegmail.com

taylor_hill@fakegmail.com

foobaz@bar.com

jacob_anderson@gameofthron.es

edward_barrett@fakegmail.com

javier_smith@fakegmail.com

john_roman@fakegmail.com

indira_varma@gameofthron.es

maisie_williams@gameofthron.es

lena_headey@gameofthron.es

name

string

Margaery Tyrell

Patrick Knight

Hot Pie

Shawn McCormick

Tywin Lannister

Mary Mitchell

Victoria Sanders

Edward Barrett

April Cole

Ms. Cathy Miller

Melissa Jones

Yolanda Owen

Thomas Morris

Anthony Cline

Nichole Miller

Brenda Martin

Joshua Kent

Victor Fleming

Tyrian Lannister

Kevan Lannister

password

string

\$2b\$12\$YjMZPfrijqS6l/ZylbxlgOV402ijFcNE.6YVueL5FoWK3eE/QVgKu

\$2b\$12\$hFkSycBpMDENs/sT.5y0feifE5iluHUJmYHqH/ISkcc8l/HRzzRSq

\$2b\$12\$h25/G5rXZejwsAhn7rvYluuLzwnayZcf4yX0CV02gDh0dAI4UyVsy

\$2b\$12\$t9xIBFoIWLphJBO9VCpV4uoZDx/TOhg7N436kBe72hgDMxsGvQr3m

\$2b\$12\$6vz7wiwO.EI5RiivqlzUc./9480gbluPtXcahDxladgyC3P58XCUC

\$2b\$12\$5GAeUijkM4g3s5yBZniXxOYIAI3vLt.CPkXLCs3jifsYJrgJ37fy

\$2b\$12\$Ki.7nartO8MscZ53cV5jEuLa83JD8OKd6sgCqxird98ijmWRTRcrC

\$2b\$12\$Dcm7ncpCt0PVR4HFux4hDetCNL3E3mQg9GKqJHR/Nz/O40HCgEJzu

preferences

Document with 0 nested fields.

undefined

Exercise 1

- A. Mostra els 2 primers comentaris que hi ha en la base de dades.
- B. Quants usuaris tenim registrats?
- C. Quants cinemes hi ha en l'estat de Califòrnia?
- D. Quin va ser el primer usuari/ària en registrar-se?
- E. Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

A. Mostra els 2 primers comentaris que hi ha en la base de dades.

The screenshot shows the MongoDB Atlas web interface for a database named 'ITAcademy' and a collection named 'movie_enjoy'. The 'Aggregations' tab is selected, and a pipeline is being built. The pipeline consists of a single stage: '\$limit' with a value of 2. The pipeline is named 'Limit'. The 'Preview' button is highlighted, and the 'PIPELINE OUTPUT PREVIEW' section shows a sample of 2 documents. The documents are JSON objects with fields: '_id', 'name', 'email', 'movie_id', 'text', and 'date'.

```
1 [
2   {
3     $limit:
4       /**
5        * Provide the number of documents to limit
6        */
7     2
8   }
9 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 2 documents

OUTPUT OPTIONS

```
{
  "_id": "ObjectId('5a9427648b0beeb69579d3')",
  "name": "Cameron Duran",
  "email": "cameron_duran@fakegmail.com",
  "movie_id": "ObjectId('573a1390f29313caabcd4217')",
  "text": "Quasi dicta culpa asperiores quaerat perferendis neque. Est animi pari...",
  "date": "1983-04-27T20:39:15.000+00:00"
}
```

```
{
  "_id": "ObjectId('5a9427648b0beeb69579ea')",
  "name": "Cameron Duran",
  "email": "cameron_duran@fakegmail.com",
  "movie_id": "ObjectId('573a1390f29313caabcd433d')",
  "text": "Ad asperiores mollitia aperiam non incidunt. Totam fugiat cumque praes...",
  "date": "1989-01-21T14:20:47.000+00:00"
}
```

B. Quants usuaris tenim registrats?

Registered users = 185 users

The screenshot shows the MongoDB Compass interface for the 'users' collection. The 'Aggregations' tab is selected, and a single stage named '\$count' is added. The pipeline is defined as follows:

```
1 [
2   {
3     $count:
4       /**
5        * Provide the field name for the count.
6        */
7       "registered_users"
8   }
9 ]
```

The 'PIPELINE OUTPUT PREVIEW' on the right shows a sample of 1 document with the result: `registered_users : 185`.

C. Quants cinemes hi ha en l'estat de Califòrnia?

California theaters = 169 theaters

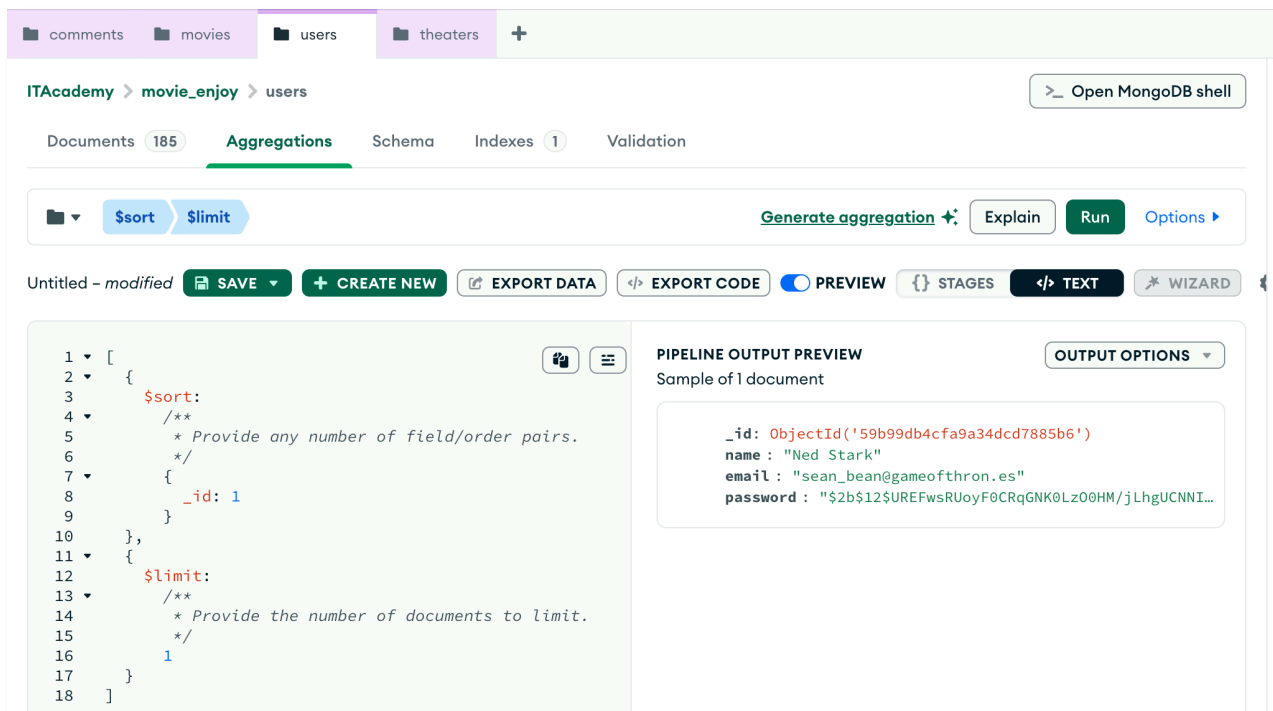
The screenshot shows the MongoDB Compass interface for the 'theaters' collection. The 'Aggregations' tab is selected, and two stages are added: '\$match' and '\$count'. The pipeline is defined as follows:

```
1 [
2   {
3     $match:
4       /**
5        * query: The query in MQL.
6        */
7       {
8         "location.address.state": "CA"
9       }
10  },
11  {
12    $count:
13      /**
14       * Provide the field name for the count.
15       */
16      "California_Theaters"
17  }
18 ]
```

The 'PIPELINE OUTPUT PREVIEW' on the right shows a sample of 1 document with the result: `California_Theaters : 169`.

D. Quin va ser el primer usuari/ària en registrar-se?

The first registered user was: Ned Stark



ITAcademy > movie_enjoy > users

Documents 185 Aggregations Schema Indexes 1 Validation

Sort Limit Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES TEXT WIZARD

```
1 [
2   {
3     $sort:
4     /**
5      * Provide any number of field/order pairs.
6      */
7     {
8       _id: 1
9     }
10  },
11  {
12    $limit:
13    /**
14     * Provide the number of documents to limit.
15     */
16    1
17  }
18 ]
```

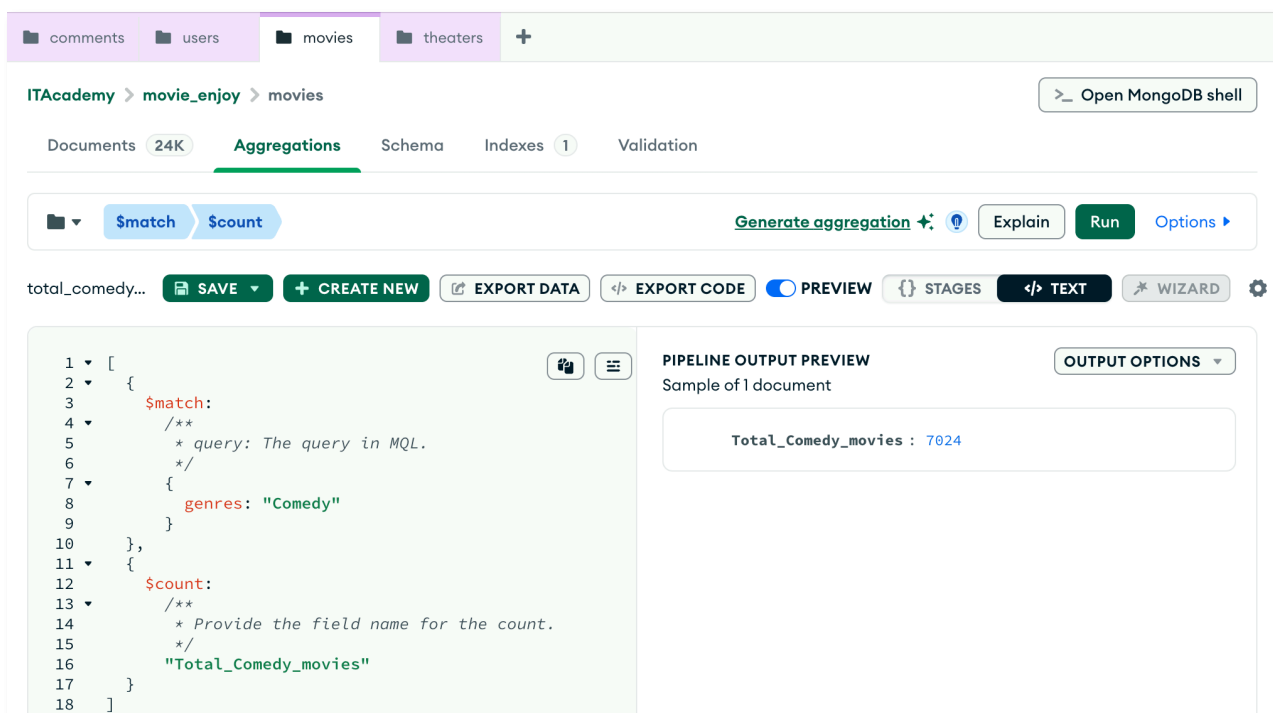
PIPELINE OUTPUT PREVIEW

Sample of 1 document

```
_id: ObjectId('59b99db4cfa9a34dcd7885b6')
name: "Ned Stark"
email: "sean_bean@gameofthron.es"
password: "$2b$12$UREFwsRUoyF0CRqGNK0Lz00HM/jLhgUCNNI..."
```

D. Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

Total Comedy movies = 7024 movies



ITAcademy > movie_enjoy > movies

Documents 24K Aggregations Schema Indexes 1 Validation

Match Count Generate aggregation Explain Run Options

total_comedy... SAVE CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES TEXT WIZARD

```
1 [
2   {
3     $match:
4     /**
5      * query: The query in MQL.
6      */
7     {
8       genres: "Comedy"
9     }
10  },
11  {
12    $count:
13    /**
14     * Provide the field name for the count.
15     */
16    "Total_Comedy_movies"
17  }
18 ]
```

PIPELINE OUTPUT PREVIEW

Sample of 1 document

```
Total_Comedy_movies : 7024
```

Exercise 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

Showing 1 of the 18th movies produced in 1932 but that the genre is drama or eaten in French.

ITAcademy > movie_enjoy > movies

Documents 24K Aggregations Schema Indexes 1 Validation

\$match

Generate aggregation Explain Run Options

Untitled - modified SAVE CREATE NEW EXPORT DATA EXPORT CODE PREVIEW STAGES TEXT WIZARD

```
1 [
2   {
3     $match:
4       /**
5        * query: The query in MQL.
6        */
7       {
8         year: 1932,
9         $or: [
10          {
11            genres: "Drama"
12          },
13          {
14            languages: "French"
15          }
16        ]
17      }
18    ]
19  ]
```

PIPELINE OUTPUT PREVIEW

Sample of 10 documents

OUTPUT OPTIONS

```
_id: ObjectId('573a1391f29313caabcd9458')
plot: "A young artist draws a face at a canvas on his easel. Suddenly the mou..."
runtime: 55
rated: "UNRATED"
cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWYjYjYi00NjA4LTk4ZW..."
title: "The Blood of a Poet"
lastupdated: "2015-09-16 13:13:05.537000000"
languages: Array (1)
released: 2010-05-20T00:00:00.000+00:00
directors: Array (1)
writers: Array (1)
awards: Object
year: 1932
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object
```

ITAcademy > movie_enjoy > movies

Documents 24K Aggregations Schema Indexes 1 Validation

\$match Edit

ALL RESULTS EXPORT DATA EXPORT CODE

Showing 1 - 18 of 18

```
_id: ObjectId('573a1391f29313caabcd9458')
plot: "A young artist draws a face at a canvas on his easel. Suddenly the mou..."
runtime: 55
rated: "UNRATED"
cast: Array (4)
num_mflix_comments: 1
poster: "https://m.media-amazon.com/images/M/MV5BYWY3ODE5ZWYjYjYi00NjA4LTk4ZW..."
title: "The Blood of a Poet"
lastupdated: "2015-09-16 13:13:05.537000000"
languages: Array (1)
released: 2010-05-20T00:00:00.000+00:00
directors: Array (1)
writers: Array (1)
awards: Object
year: 1932
imdb: Object
countries: Array (1)
type: "movie"
tomatoes: Object
```

Exercise 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

Showing 1 of the 166 movies from USA with between 5 and 9 awards and produced between 2012 and 2014.

The screenshot shows the MongoDB Atlas interface for the 'movie_enjoy' database. The 'Aggregations' tab is selected, displaying a pipeline query in the editor. The query filters for movies from the USA with between 5 and 9 awards, produced between 2012 and 2014. The 'PIPELINE OUTPUT PREVIEW' on the right shows a sample of 10 documents, with the first document highlighted.

```
1 [
2   {
3     $match:
4     /**
5      * query: The query in MQL.
6      */
7     {
8       countries: "USA",
9       "awards.wins": {
10        $gte: 5,
11        $lte: 9
12      },
13       year: {
14        $gte: 2012,
15        $lte: 2014
16      }
17     }
18   }
19 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 10 documents

```
{
  "_id": ObjectId('573a13acf29313caabd29366'),
  "fullplot": "The manager of the negative assets sector of Life magazine, Walter Mit...",
  "imdb": Object,
  "year": 2013,
  "plot": "When his job along with that of his co-worker are threatened, Walter t...",
  "genres": Array (3),
  "rated": "PG",
  "metacritic": 54,
  "title": "The Secret Life of Walter Mitty",
  "lastupdated": "2015-08-31 00:10:51.747000000",
  "languages": Array (3),
  "writers": Array (3),
  "type": "movie"
}
```

The screenshot shows the MongoDB Atlas interface for the 'movie_enjoy' database. The 'Aggregations' tab is selected, displaying the full results of the query. The 'Showing 1 - 20 of 166' indicator is circled in red. The first document is expanded, showing its full structure.

ALL RESULTS [EXPORT DATA](#) [EXPORT CODE](#) **Showing 1 - 20 of 166**

```
{
  "_id": ObjectId('573a13acf29313caabd29366'),
  "fullplot": "The manager of the negative assets sector of Life magazine, Walter Mit...",
  "imdb": Object,
  "year": 2013,
  "plot": "When his job along with that of his co-worker are threatened, Walter t...",
  "genres": Array (3),
  "rated": "PG",
  "metacritic": 54,
  "title": "The Secret Life of Walter Mitty",
  "lastupdated": "2015-08-31 00:10:51.747000000",
  "languages": Array (3),
  "writers": Array (3),
  "type": "movie",
  "tomatoes": Object,
  "poster": "https://m.media-amazon.com/images/M/MV5BODYwNDYxNDk1Nl5BMl5BanBnXkFtZT...",
  "num_mflix_comments": 1,
  "released": 2013-12-25T00:00:00.000+00:00,
  "awards": Object,
  "countries": Array (2),
  "cast": Array (4),
  "directors": Array (1),
  "runtime": 114
}
```

Level 2

Exercise 1

Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

The number of comments that users have written with the domain of their email "gameofthron.es" is 22841 comments.

Keep in mind that there are a total of 83 users with the domain "gameofthron.es".

The screenshot shows the MongoDB Atlas web interface for the 'ITAcademy' database, specifically the 'movie_enjoy' collection. The 'Aggregations' tab is active, showing a pipeline with two stages: '\$match' and '\$count'. The '\$match' stage has a query that filters comments by email domain: `{ email: { $regex: '@gameofthron.es' } }`. The '\$count' stage counts the number of documents matching the criteria, resulting in `num_comments: 22841`. The interface includes a 'PIPELINE OUTPUT PREVIEW' section showing a sample of the output document.

```
1 [
2   {
3     $match:
4     /**
5      * query: The query in MQL.
6      */
7     {
8       email: {
9         $regex: "@gameofthron.es"
10      }
11    },
12  },
13  {
14    $count:
15    /**
16     * expression: Grouping expression or string.
17     */
18    "num_comments"
19  }
20 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 1 document

```
{
  "num_comments": 22841
}
```

Exercise 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

There is only one theater in each of the three Washington D.C. postal codes.

The screenshot shows the MongoDB Atlas interface for the 'theaters' collection in the 'movie_enjoy' database. The 'Aggregations' tab is active, showing a pipeline with two stages: '\$match' and '\$group'.

Aggregation Pipeline:

```
1 [
2   {
3     $match:
4       /**
5        * query: The query in MQL.
6        */
7     {
8       "location.address.state": "DC"
9     }
10  },
11  {
12    $group:
13      /**
14       * _id: The id of the group.
15       * fieldN: The first field name.
16       */
17    {
18      _id: "$location.address.zipcode",
19      total_theaters_zipcode: {
20        $sum: 1
21      }
22    }
23  }
24 ]
```

PIPELINE OUTPUT PREVIEW
Sample of 3 documents

_id: "20016"	total_theaters_zipcode : 1
_id: "20010"	total_theaters_zipcode : 1
_id: "20002"	total_theaters_zipcode : 1

Level 3

Exercise 1

Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

John Landis has directed 4 films that have IMDb scores between 7.5 and 8.

The screenshot shows the MongoDB Atlas web interface. The top navigation bar includes tabs for 'comments', 'movies', 'users', and 'theaters'. The current view is 'ITAcademy > movie_enjoy > movies'. Below the navigation bar, there are tabs for 'Documents' (24K), 'Aggregations' (selected), 'Schema', 'Indexes' (1), and 'Validation'. The 'Aggregations' tab is active, showing a pipeline named '\$match'. The pipeline is defined as follows:

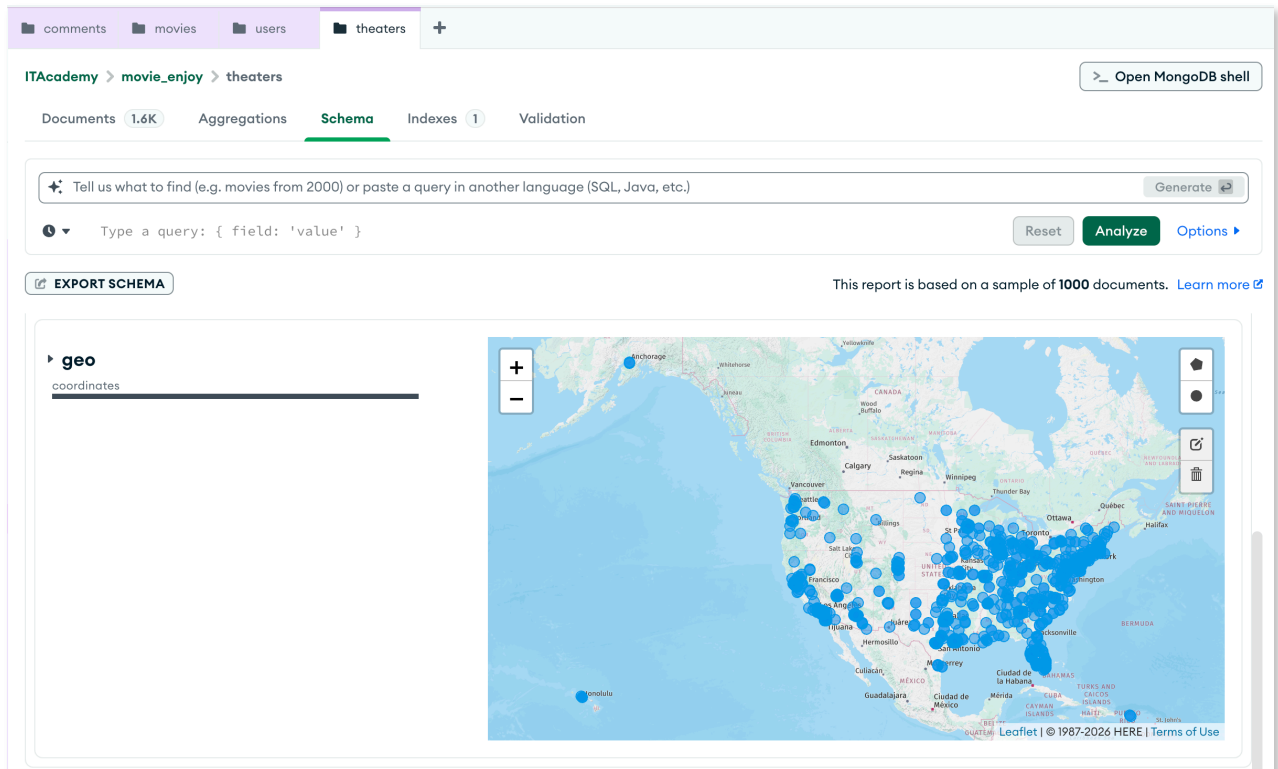
```
1 [
2   {
3     $match:
4     /**
5      * query: The query in MQL.
6      */
7     {
8       directors: "John Landis",
9       "imdb.rating": {
10        $gte: 7.5,
11        $lte: 8
12      }
13    }
14  ]
15 ]
```

Below the pipeline definition, there are buttons for 'SAVE', 'CREATE NEW', 'EXPORT DATA', 'EXPORT CODE', 'PREVIEW' (selected), 'STAGES', 'TEXT', 'WIZARD', and a settings gear icon. The 'PREVIEW' button is active, showing a 'PIPELINE OUTPUT PREVIEW' section. This section displays a sample of 4 documents, with the first document expanded to show its details:

```
{
  "_id": ObjectId('573a1397f29313caabce6d94'),
  "fullplot": "Faber College has one frat house so disreputable it will take anyone. ...",
  "imdb": {
    "year": 1978,
    "plot": "At a 1962 college, Dean Vernon Wormer is determined to expel the entire...",
    "genres": Array (1),
    "rated": "R",
    "metacritic": 82,
    "title": "Animal House",
    "lastupdated": "2015-09-13 00:02:47.803000000",
    "languages": Array (2),
    "writers": Array (3),
    "type": "movie"
  }
}
```


Exercise 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.



EXTRA learning related to the agenda outside of Task -5

Power BI connection with MongoDB:

Due to my Hardware configuration I have experimented with the Power BI connection and the MongoDM database:

- My Hardware is a Macbook Air M4 16GB

Problem:

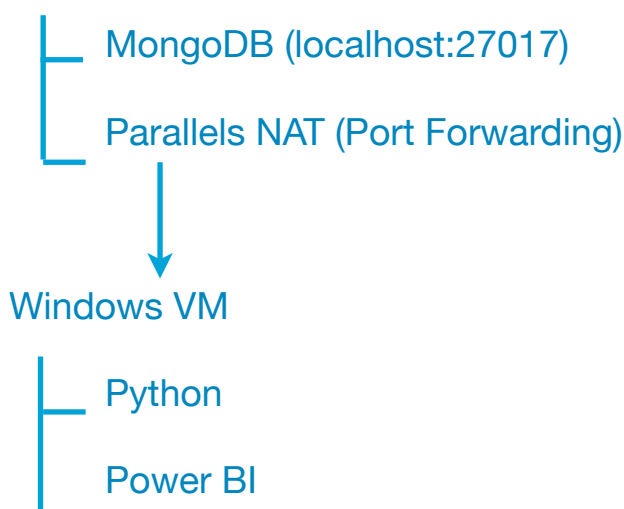
- Power BI does not currently have a local version for OSX operating system.

Solution:

- Install the MongoDB database on the Macbook's native OSX operating system.
- Create Windows 11 virtual machine (ARM) with Power BI Installation.

After several attempts to implement solutions using the Mongo BI connector, I discovered system incompatibility, other alternatives were tested and finally opted for the option of using Python Script in Power BI to connect with Mongo DB. Logically it has been necessary to configure a PortForwarding in the Parallels connection and use MongoClient from the Python pymongo library.

Mac



Python script that I run in the Power BI - Power Shell, to connect with MongoDB from Windows, and import and/or update the data in PowerBI:

```
from pymongo import MongoClient
```

```
import pandas as pd
```

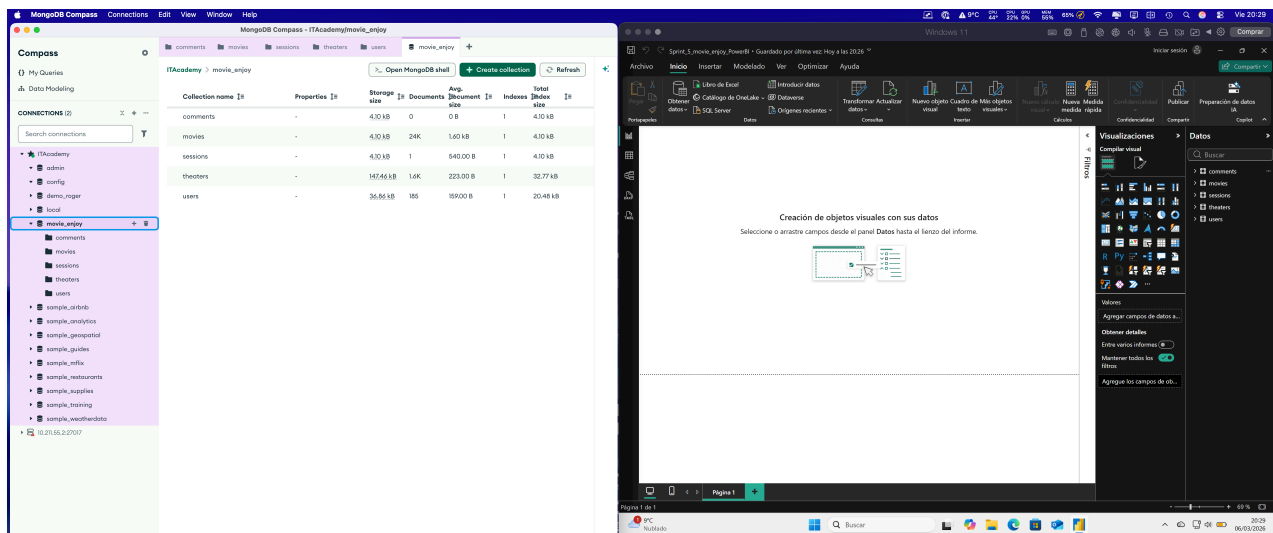
```
client = MongoClient("mongodb://10.211.55.2:27017")
```

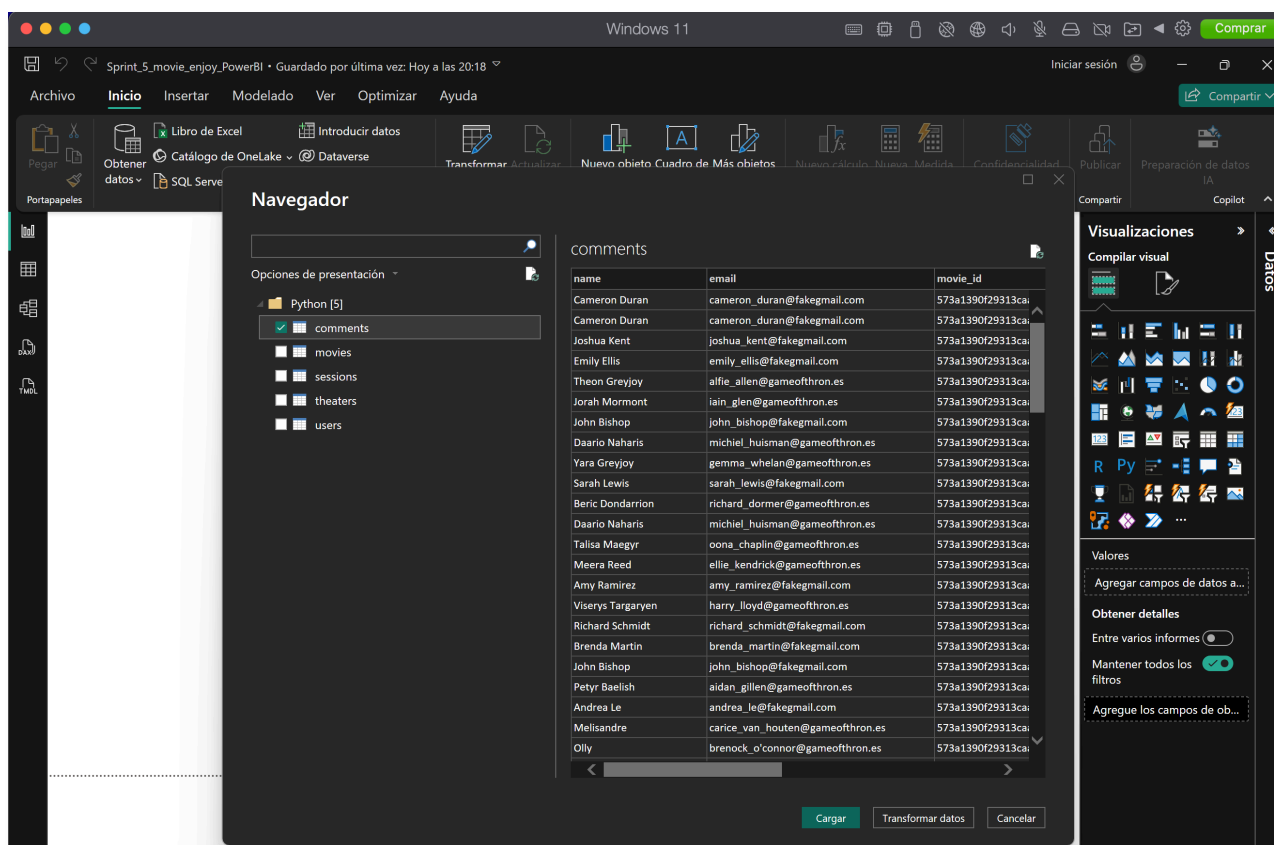
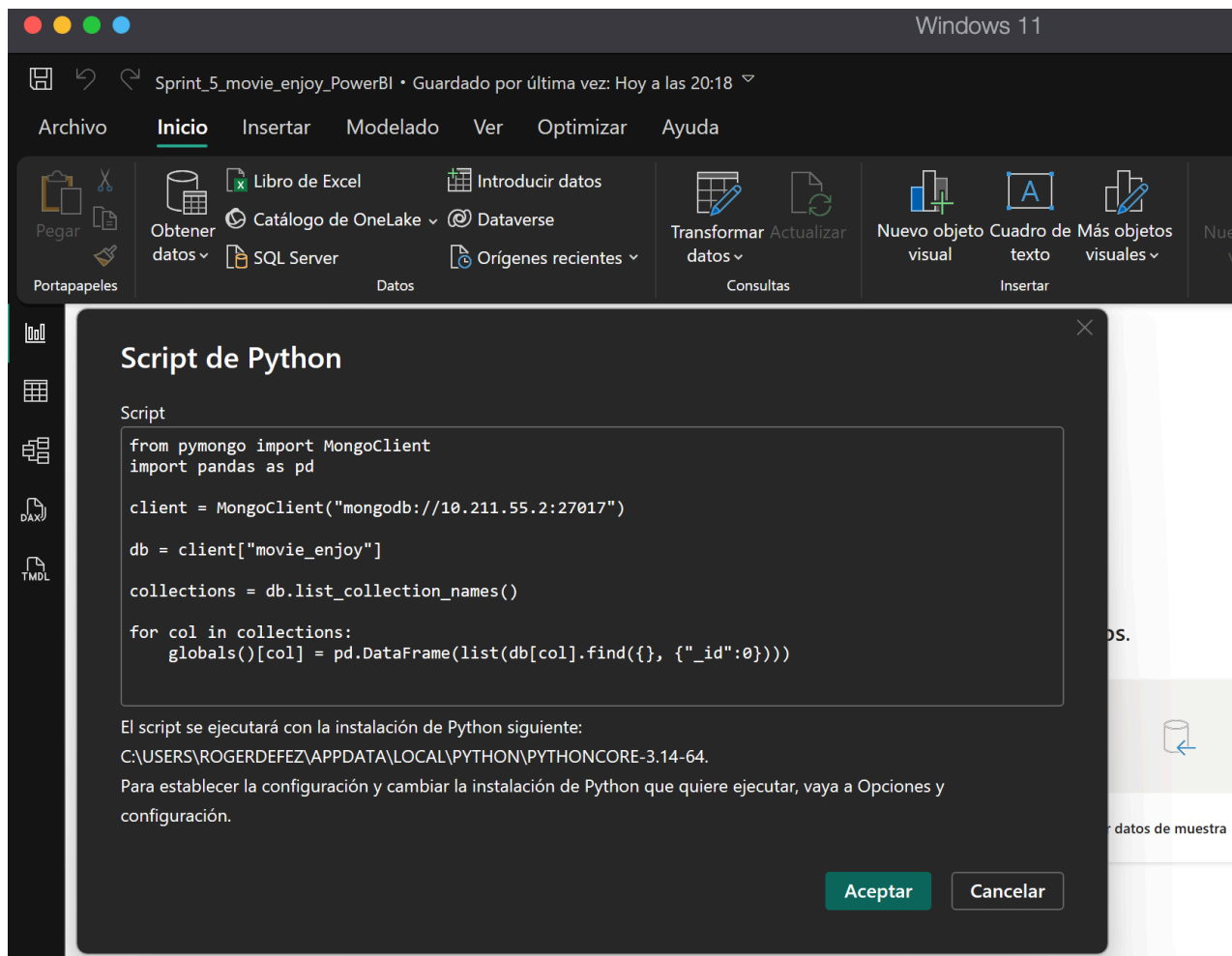
```
db = client["movie_enjoy"]
```

```
collections = db.list_collection_names ()
```

```
for col in collections:
```

```
globals () [col] = pd.DataFrame (list (db[col] find({}, {"_id":0})))
```





Windows 11

Sprint_5_movie_enjoy_PowerBI • Guardado por última vez: Hoy a las 20:18

Inicio Transformar Agregar columna Vista Herramientas Ayuda

Cerrar y aplicar Cerrar

Nuevo origen Orígenes recientes Nueva consulta

Especificar datos Configuración de origen de datos Administrar parámetros Exportar los resultados de la consulta Actualizar vista previa Consultas

Propiedades Editor avanzado Administrar

Elegir columnas Quitar columnas Conservar filas Quitar filas Reducir filas Ordenar

Dividir columnas

Consultas [5]

comments movies sessions theaters users

Table.TransformColumnTypes(users1,{{"name", type text}, {"email", type text}, {"password", type text}}

A#	name	email	password
1	Jon Snow	kit_harington@gameofthron.es	\$2b\$12\$FDeu1Ru6tLWAVidMN.b0.929BIfnyqdGuhWMYzfoA/A...
2	Sansa Stark	sophie_turner@gameofthron.es	\$2b\$12\$SnCIVE81...AtAoyPZk19.G5V0Edlwsw2h1F18kWEr3dlpG/...
3	Arya Stark	maise_williams@gameofthron.es	\$2b\$12\$19h3KjeTg3/pPNFhtdb6eGWK-CxWVXVSh9P0S5P59K5...
4	Robb Stark	richard_madden@gameofthron.es	\$2b\$12\$XPLVWQW7tjWc/PX9JMVnRn08w.IR6hv144ee8pc8Nd5Wl...
5	Theon Greyjoy	alfie_allen@gameofthron.es	\$2b\$12\$5x574mziridS3mEQVTbKbY.IK.ngDyZInT1767Gk6n4InW...
6	Ned Stark	sean_bean@gameofthron.es	\$2b\$12\$SUREFwRUYoFPCrGqGNKOLzOOHM/JLhgUCNNI9RUJqMU...
7	Daenerys Targaryen	emilia_clarke@gameofthron.es	\$2b\$12\$NzpbWHDMyemLttFFKduHenZ2N.rvxlKuYMAAWLTfaU...
8	Bran Stark	isaac_hempstead_wright@gameofthron.es	\$2b\$12\$Z7/zVm8eWMDWtIg.doS.UO7JbsBA9ibLomND1VxiZdA...
9	Joffrey Baratheon	jack_gleeson@gameofthron.es	\$2b\$12\$NVLjRg5tZCtCay30.MILO4IRblHslHfWYtUpF8bvnf954O...
10	Sandor Clegane	rory_mccann@gameofthron.es	\$2b\$12\$SIKIKOBMBN76vskRy1AjuELigfmx.SeG8vu73k24fMcO6...
11	Tyrian Lannister	peter_dinklage@gameofthron.es	\$2b\$12\$5tHwQNXyIQzP2REobUDlzuQimjzTx1GnwP.xkFULEu...
12	Khal Drogo	jason_momoa@gameofthron.es	\$2b\$12\$7tgpVkbXUqQYfKHj2yoMuEtF5BSL.FYkOXHu4zCRLrs1...
13	Samwell Tarly	john_bradley-west@gameofthron.es	\$2b\$12\$Zcz9wkjIT/dCCmupizZ...JA3:c9HoufxRygl1DKY9hsCIT0iYf...
14	Davos Seaworth	liam_cunningham@gameofthron.es	\$2b\$12\$5jbgNoWGS7LHlNm4axwXDz.tkFITsmw/aylly/IZdajRgnH2...
15	Cersei Lannister	lena_headey@gameofthron.es	\$2b\$12\$Fexjgr7CLHnCa.oU8B9seub8mqcH2uCFZ8heMc8CeK0Zf...
16	Petyr Baelish	aidan_gillen@gameofthron.es	\$2b\$12\$5qM.YvmiekyYYY7p7phpK3OicbRCDKn7ESwYAnG/o9YnFh...
17	Jeor Mormont	james_cosmo@gameofthron.es	\$2b\$12\$5z5d5vAjduFuuxFQwMH2QJwb3QKx9/gR5uefaQ3gHoj...
18	Bronn	jerome_flynn@gameofthron.es	\$2b\$12\$9zrMIF3lv7kAsd7dam2eDOJWDSN.qi2hpY5tyOFDPe545...
19	Melisandre	carice_van_houten@gameofthron.es	\$2b\$12\$2d5vCDMwVLFgJGD0sDbFLOD/utCnkbV00VJ8IUHq5K...
20	Jorah Mormont	lain_glen@gameofthron.es	\$2b\$12\$5K8BkwnprkrsBPzAS2xO/yj79kvupiV0GjA3X106AKx3...
21	Stannis Baratheon	stephen_dillane@gameofthron.es	\$2b\$12\$5bPwOM9Qk5OxORgLSZiqe9JPjBx1Uwm06d3YVVKx...
22	Margaery Tyrell	natalie_dormer@gameofthron.es	\$2b\$12\$5AJC.876rd2dgsSlu.5c4qembdVfdDajMigueMYuOBoskJUG...
23	Catelyn Stark	michelle_fairley@gameofthron.es	\$2b\$12\$5iaTH55h1zKNFXZ/FTEreWGjoxjxvmV7XL.qfQcR8CwOxK...
24	Viserys Targaryen	harry_loyd@gameofthron.es	\$2b\$12\$5cpwVmU4DyucgwpdrVJhaudzbK0XIHrBf.tpCuhJpAqon...
25	Robert Baratheon	mark_addy@gameofthron.es	\$2b\$12\$5yGqxlG9L2pXa2xVDhuPnS0Zd.VURVlz7wgOLY3pn0Os7...
26	Jaime Lannister	nikolaj_coster-waldau@gameofthron.es	\$2b\$12\$56vz7wiwO.Ei5RIrlvq1zUc./9480gb1uPlXcahDxladyC3PS8...
27	Varys	conleth_hill@gameofthron.es	\$2b\$12\$5mV9NHXLISU3fDdd5tU2.6D5tBgnC6iWQQR.1h/MsD2p...

4 COLUMNAS, 185 FILAS Generación de perfiles de columnas basada en las 1000 primeras filas

VISTA PREVIA DESCARGADA A LAS 20:12

Página 1 de 1

9°C Nublado

20:25 06/03/2026

Windows 11

Sprint_5_movie_enjoy_PowerBI • Guardado por última vez: Hoy a las 20:26

Guardar Inicio Insertar Modelado Ver Optimizar Ayuda

Obtener datos Libro de Excel Introducir datos

Catálogo de OneLake Dataverse

Transformar Actualizar datos

Nuevo objeto Cuadro de Más objetos visuales

Nuevo cálculo Nueva medida

Confidencialidad Publicar Preparación de datos IA

Portapapeles Datos Orígenes recientes Consultas Cálculos

Visualizaciones

Compilar visual

Filtros

Valores

Agregar campos de datos a...

Obtener detalles

Entre varios informes

Mantener todos los filtros

Agregue los campos de ob...

Creación de objetos visuales con sus datos

Seleccione o arrastre campos desde el panel Datos hasta el lienzo del informe.

Página 1 de 1

9°C Nublado

20:28 06/03/2026

